STOCHASTIC METHODS IN WATER RESOURCES

Unit 1: Introduction to probability and statistics
Lecture 4a: Model estimation and testing

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Generalities

- Statistical inference deals with statistical estimations based on a sample from the population.
- Some definitions:
  - **Population**: consist of all possible observations of a process (e.g. air temperature at certain location). Some of the observations in the in the population may not have any physical sense, perhabs, due to sensor errors.
  - **Sample**: is a subset of the population (e.g. instantaneous daily streamflow for a certain period in a station). A random sample is thus a sample that is representative of the population.
  - **Random variables**: is a real-valued function defined on a sample space. Wheather a random variable is discrete or continuous depends on how the sample space is defined.
  - **Statistic**: is a function of the observations that is quantifiable and does not contain any unknown parameter. Note that a statistic is also a random variable that provides an estimation.
  - **Estimator**: is the method or rule of estimation. For instance, the sample mean $\bar{X}$ is a point estimator of the population mean μ.
  - **Estimate**: is the value yielded by the estimator.
- Suppose that the population of variable X follows a Normal distribution and the distribution parameters θ are unknown. Thus, a random sample of X of size n.
- Parameters θ can be described by a number or a range; this last include an uncertainty.

Properties of estimators

Unbiasedness

Given a sample of observations of a random variable X, $X_1, X_2, \cdots, X_n$, the objective is to estimate the value of the parameter θ, which is also a random variable. The idea is to seek for a estimator of θ that get as closest as possible to the true value. This estimator will produce statistics that are distributed following a certain pdf. Following this, a **point estimator $\hat{\theta}$** is an **unbiased estimator** of the population parameter θ if $E[\hat{\theta}] = \theta$. If the estimator is biased, the bias $= E[\hat{\theta}] - \theta$.

Mean of the sample mean

Let us show that the sample mean $\bar{X}$ is an unbiased estimator of μ. If $\bar{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$, then $E[\bar{X}] = \frac{1}{n} nE[X_i] = \frac{1}{n}(n\mu) = \mu$.

Note that many estimator are biased and methods such as jackknife and bootstrap are used to reduce the bias. Ideal estimator must also have the following properties.

Consistency

An estimator $\hat{\theta}_n$, based on a sample size n, is a **consistent** estimator of θ if for any positive number ε $\lim_{n\to\infty} Pr[|\hat{\theta}_n - \theta| \leq \varepsilon] = 1$

Minimum variance

A part for looking for an unbiased estimator, it is also desirable to seek for an **minimum variance estimator**. The minimum variance unbiased estimator is the estimator with the smallest variance out of all unbiased estimator

Properties of estimators

Efficiency

Efficiency is the relative measure of the variance of the sampling distribution, with the efficiency increasing as the variance decrease. In terms of the mean square error (mse) an estimator with the minimum *mse* among all possible unbiased estimators is called and **efficient estimator**. If A is an estimator of θ, the *mse* is:

$$\begin{aligned} E\left[(A-\theta)^2\right] &= E\left[((A-E[A])-(\theta-E[A]))^2\right] \\ &= E[(A-E[A])^2]+(\theta-E[A])^2 \\ &= Var[A]-bias^2 \end{aligned}$$

The *mse* of an estimator can be used as a relative measure of efficiency when comparing two or more estimator.

Sufficiency

So far, properties such as unbiasedness, consistency and the minimum *mse* are key to select the most suitable estimators. Thus, a **sufficient estimator** provides as much information as possible about a sample of observations of a random variable X. Let a sample $X_1, X_2, \cdots, X_n$ be drawn randomly from population having probability distribution with unknown parameters θ. Thus, the statistic $T = f(X_1, X_2, \cdots, X_n)$ is said to be sufficient to estimate θ if the distribution of $X_1, X_2, \cdots, X_n$ conditional to the statistics T is independent of θ.

Median and mean

The median, taken as a measure of mean density or central tendency, does not contain all the information in a sample. The median is the middle value of the sample; if any other value is changed the mean changes but the median is unaltered.

Estimation of confidence intervals

The uncertainty of point estimates can be quantified by the relative variances or mean square errors of the estimators. Because of this uncertainty, the next step of inference is **interval estimation**. Two numbers, say, a and b, that are expected to include within their range an unknown parameter θ in a specified percentage of cases after repeated experimentation under identical conditions. That is, in place of one statistic that estimates θ, we find a range specified by two statistics, which includes it at a given level of probability. The end points a and b of this range are known as **confidence limits**, and the interval (a, b) is known as the **confidence interval**. We do not have the precision as for a point estimator but we have confidence (without absolute certainty) that the assertion is right.

Confidence interval estimation of the mean when the standard deviation is known

Let C_l and C_u be the lower and upper confidence limits that include an unknown but invariable parameter θ. Although there is some uncertainty associated with this statement, we will be right in a proportion, say, $1-\alpha$, of the samples taken from the same population, on average, when we make the assertion that the given interval includes θ. Thus we can say, by adopting the long-run frequency interpretation of probability,

$$Pr[C_l \leq \theta \leq C_u] = 1 - \alpha$$

where θ is a constant, but the estimator $\hat{\theta}$ and the confident limits C_l and C_u are random variables. Recall that examples of $\hat{\theta}$ are $\bar{X}$ and $\bar{S}^2$. The probability $(1-\alpha)$ is known as the confidence level or confidence coefficient. It often takes values such as 0.99, 0.95, and 0.90. The confidence limits C_l and C_u depend on the sampling distribution of θ. The standard deviation of the statistic $\hat{\theta}$ called its standard error.

Estimation of confidence intervals

Confidence interval estimation of the mean when the standard deviation is known

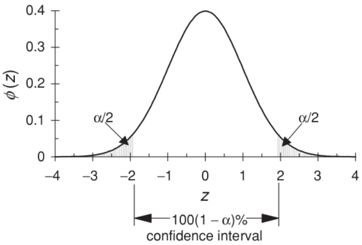
In some situations, we may require one-sided confidence limits. The lower and upper one-sided confidence limits are specified, respectively, by

$$Pr[C_l \leq \theta] = 1 - \alpha \quad \text{for } 0 < \alpha < 1$$

and

$$Pr[\theta \leq C_u] = 1 - \alpha \quad \text{for } 0 < \alpha < 1$$

Let $\bar{X}$ be the mean of a random sample of size n drawn from a population with known standard deviation σ. The 100(1–α) percent central two-sided confidence interval for the population mean μ is $\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$, where $z_{\alpha/2}$ is a standard normal variate that is exceeded with probability $\alpha/2$, $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \quad N \sim (0,1)$. The one-sided upper and lower 100(1–α) percent confidence limits for the population mean μ are, $\bar{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}$ and $\bar{X} - z_{\alpha}\frac{\sigma}{\sqrt{n}}$, respectively.



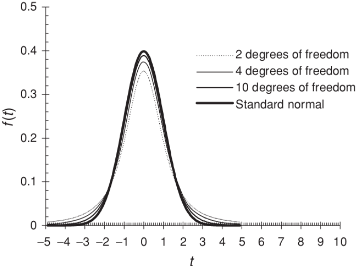
Estimation of confidence intervals

Confidence interval estimation of the mean when the standard deviation is unknown

Quite often in practice the mean and standard deviation are both unknown. Under such conditions we must modify our approach. We assume normality of the variate X as before, but the consequences of a nonnormal distribution are minor for small to moderate departures from normality, if the sample size is large, say, $n > 30$. In this situation we apply the Student's t distribution.The T variable represents the mean $\bar{X}$ standardized as $T = \frac{\bar{X}-\mu}{\hat{S}/\sqrt{n}} \sim t_{n-1}$. Note that T has a Student's t distribution with $\nu = (n-1)$ degrees of freedom. Recall that the Student's t distribution pdf is:

$$f_T(t) = \frac{\Gamma[(\nu+1)/2]}{\sqrt{\pi\nu}\Gamma(\nu/2)} \frac{1}{[(t^2/\nu)+1]^{\frac{\nu+1}{2}}} \quad \text{for} \quad -\infty < t < \infty$$

Let $\bar{X}$ and $\hat{S}$ be the mean and the standard deviation, respectively, of a random sample of size n drawn from a normal distribution with unknown standard deviation σ. The $100(1-\alpha)$ percent central two-sided confidence interval for the population mean μ is $\left(\bar{X} - t_{n-1,\alpha/2}\frac{\hat{S}}{\sqrt{n}}, \bar{X} + t_{n-1,\alpha/2}\frac{\hat{S}}{\sqrt{n}}\right)$, where $t_{n-1,\alpha/2}$ is Student's t variate with $n-1$ degrees of freedom and probability of $\alpha/2$ of execeedance.



Confidence interval for a proportion

Consider a two-sided Bernoulli variable that occur with probability p if it is success and $1 - p$ if it is failure. Recall that mean and the variance of this variable are p and $p(1 - p)$, respectively. If an experiment involving a Bernoulli variable is repeated n times, the standard error of the estimated proportion of a success is:

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

Note that for a large sample size, for instance, $n > 30$, and for $np > 5$ and $n(1 - p) > 5$. the sampling distribution is nearly normal.

Estimation of confidence intervals

Sampling distribution of differences and sums of statistics

Let S_1 and S_2 independent statistics from two populations, computed based on sample sizes n_1 and n_2, respectively. Let the means and the standard errors of the sampling distributions of the two statistics be μ_{S_1}, μ_{S_2} and σ_{S_1}, σ_{S_2}, respectively. The differences between the statistics (after repeated sampling) from the two populations have a sampling distribution with mean $\mu_{S_1 - S_2} = \mu_{S_1} - \mu_{S_2}$, and standard error $\sigma_{S_1 - S_2} = \sqrt{\sigma_{S_1}^2 + \sigma_{S_2}^2}$. The sampling distribution of the sum of the statistics $S_1 + S_2$ has a mean $\mu_{S_1 + S_2} = \mu_{S_1} + \mu_{S_2}$, and standard error $\sigma_{S_1 + S_2} = \sqrt{\sigma_{S_1}^2 + \sigma_{S_2}^2}$.

Confidence limits for differences between two means of the annual rainfalls at two stations

At station 1, annual rainfall has been measured over $n_1 = 50$ years where $\bar{x_1} = 900$ mm and $\hat{s}_1 = 80$ mm. At station 2, annual rainfall has been measured over $n_1 = 40$ years where $\bar{x_1} = 825$ mm and $\hat{s}_1 = 90$ mm. Because annual rainfalls are the result of a large number of small causes, such an additive effect makes it plausible to assume that the distribution is normal. Thus, the sampling distribution of the difference between the two means is also normal.

The $(1 - \alpha)$ percent confidence limits for the difference between the two means are found using:

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

where $z_{\alpha/2}$ is the standard normal variate which is exceeded with probability $\alpha/2$.

Replacing $\hat{s_1}$ for σ_1 and $\hat{s_2}$ for σ_2, the 95% confident limits are:

$$75 + 1.96\sqrt{\frac{80^2}{50} + \frac{90^2}{40}} = 75 \pm 36mm$$

limits of $(39mm, 111mm)$.

Estimation of confidence intervals

Interval estimation for the variance: chi-squared distribution

Consider a set of normally and independently distributed random variates X_i where $i = 1, 2, \cdots, \nu$, with means μ_i and variances σ_i^2. If we standardize the X_i by subtracting the mean and dividing by the standard deviation, respectively, to a set Z_i, $i = 1, 2, \cdots, \nu$ with zero mean and unit variance. Consider the random variable formed by the sum of squares of the Z variables, say $\chi^2 = Z_1^2 + Z_2^2 + \cdots + Z_\nu^2$. The mgf of χ^2 is given by:

$$M_{\chi^2}(t) = E[e^{t\chi^2}] = \prod_{i=1}^{\nu} E[e^{tZ_i^2}]$$

Regarding that:

$$E[e^{tZ_i^2}] = \int_{-\infty}^{\infty} e^{tz^2} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz = \frac{1}{\sqrt{1-2t}} \int_{-\infty}^{\infty} \frac{\sqrt{1-2t}}{\sqrt{2\pi}} e^{-(1-2t)z^2/2} dz$$

the integral on the right-hand side is unity, being a representation of the area under a normal curve with a mean of zero and variance $1/(1-2t)$, so:

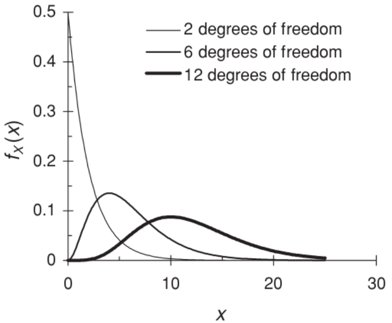
$$M_{\chi^2}(t) = \left[\frac{1/2}{1/2 - t}\right]^{\nu/2}$$

Estimation of confidence intervals

Interval estimation for the variance: chi-squared distribution

If we compare this mgf with the one for the gamma distribution, the cdf of the chi-squared distribution with ν degrees of freedom is:

$$F(\chi^2) = \frac{1}{2}\int_0^{\chi^2} \frac{(u/2)^{(\nu/2)-1} e^{-u/2}}{\Gamma(\nu/2)} du$$



Considering a random sample $X_1, X_2, \cdots, X_n$ of independent normally distributed variables with common mean μ, variance σ^2 and sample mean $\hat{X}$.

$$\hat{S}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

is an unbiased estimator of the variance σ^2. Let us now consider the quantity

$$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n}\left[(X_i - \bar{X}) + (\bar{X} - \mu)\right]^2$$

Estimation of confidence intervals

Interval estimation for the variance: chi-squared distribution

because $\sum_{i=1}^{n}(X_i - \bar{X}) = 0$, the cross-product in this equation is zero, so:

$$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n}(X_i - \bar{X})^2 + n(\bar{X} - \mu)^2$$

Dividing by σ^2 and substituting $\hat{S}^2$ from the equation above for the second term:

$$\frac{\sum_{i=1}^{n}(X_i - \mu)^2}{\sigma^2} = \frac{(n-1)\hat{S}^2}{\sigma^2} + \frac{(\bar{X} - \mu)^2}{\sigma^2/n}$$

If the first term of this equation is distributed as χ_n^2, and according to the Central Limit Theorem, the last term is distributed as χ_1^2. Thus, taking account of the additive property of the gamma and chi-squared variates, we can say that $\frac{(n-1)\hat{S}^2}{\sigma^2}$ has χ_{n-1}^2 distribution, that is, with $\nu = n - 1$ degrees of freedom. This important result is used in finding confidence limits for the variance σ^2 of a normal population. The chi-squared cdf is given in tables for various values of the degrees of freedom ν. For example, the (1–α) percent two-sided confidence interval for σ^2 is found from:

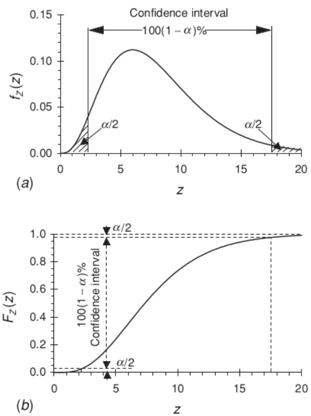
$$Pr\left[\chi_{n-1,1-\alpha/2}^2 \leq \frac{(n-1)\hat{S}^2}{\sigma^2} \leq \chi_{n-1,\alpha/2}^2\right] = 1 - \alpha$$

Estimation of confidence intervals

Interval estimation for the variance: chi-squared distribution

This probability is represented by the area between the two vertical lines of the figure which shows the pdf. These values are from left to right the values that a χ^2_{n-1} variate exceeds with probabilities $(1-\alpha/2)\%$ and $\alpha/2\%$, respectively. Taking reciprocal of the variables in the equation, the positions of the chi-squared variables need to be changed, and multiplying by $(n-1)\hat{S}^2$, we obtain:

$$Pr\left[\frac{(n-1)\hat{S}^2}{\chi^2_{n-1,\alpha/2}} \leq \sigma^2 \leq \frac{(n-1)\hat{S}^2}{\chi^2_{n-1,1-\alpha/2}}\right] = 1-\alpha$$



Hypothesis testing

- A second major area of statistical inference is the **testing of hypotheses**, which is closely related to the determination of confidence limits.
- What we are examining concerns the parameters or form of the probability distribution that yields the observations.
- This involves making a declaration or statement called a **hypothesis** about a population.
- The discussion is focused on certain assumptions (regarding, for instance, the mean of the population) without initially making any commitment on assuming them.
- From a philosophical viewpoint, our conception of the actual state of nature may not be correct. However, we have a sample of data that represents the physical reality, and although our initial hypothesis may be somewhat tentative, we can revise the concept each time new information becomes available and thus come closer to the true state of nature.
- The approach we follow is akin to the verification of some of the scientific hypotheses of scientists and engineers. For instance, it is usually feasible to ascertain whether a procedure for treating wastewater makes a change in the quality of the effluent, as measured, for example, by the biochemical oxygen demand. As another example, there may be reason to believe that the distribution of high flows in a river has changed on account of climatic effects or because the flow regime has been altered.

Hypothesis testing

- A random sample is taken from the population and **statistical hypotheses**, called **null** and **alternative**, are declared. Then a statistical test is made.
- If the observations do not support the model or theory postulated, the null hypothesis is **rejected** in favor of the alternative one, which may be considered to be true. However, if the observations are in agreement, then the null hypothesis is **not rejected**.
- This does not necessarily mean that it is accepted. It suggests that there is insufficient evidence against the null hypothesis in favor of the alternative one.
- Associated with the decision is a **level of significance**, α. This is complementary to the probability introduced for setting confidence limits.
- For instance, an engineer may question whether there has been a significant change, for instance, in a mean rate of traffic flow on a highway or an average strength of a material. On the contrary, such a variation may sometimes be due to sampling differences without any change in the mean or other aspect of the population. It is then said to be **not significant**.

Hypothesis testing

- Hypothesis testing thus concerns procedures for rejecting a statement or not rejecting it and the chances of making incorrect decisions of either kind. It also involves the use of a particular function of the sample measurements.
- If we assume that the observations come from a normal or other specified population, then the test is called **parametric**. In contrast, **nonparametric tests** are not based on such assumptions.